

DICKINSON ROOSEVELT, ND, USA

WMO: 727630

Lat: 46.799N

Lon: 102.797W

Elev: 786

StdP: 92.23

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24012

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.3	(c) -23.9	(d) -30.9	(e) 0.2	(f) -27.0	(g) -27.5	(h) 0.3	(i) -23.7	(j) 14.9	(k) -7.9	(l) 13.7	(m) -6.8	(n) 5.2	(o) 290	(p) 0.742

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	33.9	19.1	32.0	19.0	29.8	18.4	22.0	28.9	20.8	28.0	19.8	26.9	5.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	16.2	25.6	18.6	14.8	24.2	17.4	13.7	22.9	68.6	28.8	63.9	28.1	60.0	26.6	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.4	(b) 11.9	(c) 10.8	(e) -30.6	(f) 37.7	(g) 2.6	(h) 1.9	(i) -32.4	(j) 39.1	(k) -34.0	(l) 40.2	(m) -35.4	(n) 41.3	(o) -37.3	(p) 42.7		
(4) 13.4	(5) 11.9	(6) 10.8	(7) -30.8	(8) 24.1	(9) 2.5	(10) 1.3	(11) -32.6	(12) 25.1	(13) -34.1	(14) 25.9	(15) -35.5	(16) 26.6	(17) -37.4	(18) 27.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-8.7	-7.3	-1.3	5.3	11.4	16.8	21.0	20.3	14.9	6.8	-0.8	-7.0	
	DBStd	12.13	8.22	8.00	7.74	5.75	4.69	3.84	3.42	3.72	5.03	5.58	6.81	7.69	
	HDD10.0	2614	581	483	354	158	40	2	0	0	13	129	326	527	
	HDD18.3	4726	839	717	609	392	219	71	13	23	126	359	574	785	
	CDD10.0	1164	0	0	3	17	83	208	342	319	160	29	2	0	
	CDD18.3	234	0	0	0	0	4	26	97	84	22	1	0	0	
	CDH23.3	3249	0	0	2	18	94	359	1221	1129	392	33	1	0	
	CDH26.7	1318	0	0	0	3	20	102	528	504	153	8	0	0	
Wind	WSAvg	5.4	5.8	5.5	5.7	5.7	5.7	5.1	4.5	4.7	5.0	5.5	5.5	5.6	
Precipitation	PrecAvg	410	10	11	17	48	56	86	58	35	41	22	13	9	
	PrecMax	646	25	59	64	121	146	183	144	101	159	136	69	29	
	PrecMin	270	1	0	2	2	3	14	9	0	4	0	0	0	
	PrecStd	91	7	11	17	33	34	40	36	31	34	28	15	7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	12.9	21.3	25.8	29.0	32.4	37.6	36.5	34.4	27.2	19.3	11.4	
		MCWB	4.7	5.9	10.1	12.8	16.1	20.2	19.9	19.6	18.0	13.9	9.5	5.3	
	2%	DB	6.0	8.4	16.3	21.4	26.0	29.5	33.9	33.8	30.8	22.7	15.0	7.7	
		MCWB	2.4	3.6	7.6	11.0	15.4	19.0	19.5	18.8	17.1	12.6	7.2	3.1	
	5%	DB	3.8	5.2	12.7	18.0	23.4	27.3	31.9	32.0	27.8	18.9	11.3	4.8	
		MCWB	1.2	2.1	6.3	9.3	13.6	18.3	19.7	18.6	15.9	10.4	5.3	1.4	
	10%	DB	2.2	2.8	9.3	15.5	21.0	25.2	29.6	29.6	24.9	15.8	8.2	2.7	
		MCWB	0.2	0.5	4.5	8.0	13.0	16.9	19.4	18.1	15.0	9.1	3.7	0.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	6.8	10.8	13.7	18.4	23.3	24.3	22.8	20.3	15.3	10.0	5.6
			MCDB	10.0	12.0	19.2	23.5	24.9	29.4	30.2	30.4	28.9	25.0	18.8	10.7
2%		WB	2.8	4.1	8.6	11.7	16.4	20.9	22.5	21.1	18.7	13.1	7.7	3.4	
		MCDB	5.7	8.0	15.9	20.4	23.7	26.4	29.0	28.8	26.6	20.0	14.3	7.1	
5%		WB	1.4	2.2	6.6	10.0	14.9	19.4	21.4	20.1	17.3	11.4	5.7	1.7	
		MCDB	3.7	5.1	12.4	17.0	21.4	25.3	28.5	27.7	24.8	17.9	10.6	4.7	
10%		WB	0.2	0.8	4.7	8.6	13.5	18.0	20.3	19.1	15.7	9.7	4.1	0.3	
		MCDB	2.1	2.7	9.1	14.6	19.6	23.7	27.3	26.7	23.4	15.2	8.0	2.6	
Mean Daily Temperature Range		5% DB	MDBR	10.1	10.4	11.6	13.1	13.1	13.1	14.8	15.6	14.6	12.5	11.2	9.9
			MCDBR	11.1	12.9	17.2	18.3	17.4	16.1	17.9	18.7	19.3	18.0	16.1	12.4
	MCWBR		8.4	8.9	10.0	9.9	8.6	7.9	7.1	7.1	8.2	9.4	9.6	8.6	
	5% WB	MCDBR	10.8	12.2	16.6	17.1	14.9	14.6	15.1	16.0	16.8	16.4	15.4	12.2	
		MCWBR	8.4	8.6	10.0	9.5	8.0	8.1	7.5	7.2	7.7	8.8	9.4	8.8	
Clear-Sky Solar Irradiance	taub	0.252	0.258	0.276	0.317	0.339	0.350	0.359	0.368	0.325	0.295	0.263	0.248		
	taud	2.500	2.524	2.535	2.424	2.421	2.426	2.433	2.376	2.476	2.544	2.526	2.511		
	Ebn at Noon	871	938	963	941	925	914	899	876	891	876	847	835		
	Edh at Noon	62	75	87	107	111	112	109	111	92	73	60	55		
All-Sky Solar Radiation	RadAvg	1.65	2.64	3.77	4.91	5.81	6.57	6.93	5.72	4.33	2.75	1.74	1.26		
	RadStd	0.13	0.27	0.29	0.29	0.52	0.49	0.28	0.38	0.32	0.30	0.21	0.06		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page